### 8.4.4 Starting the cluster

To run the application, go to the Git repository. Launch four sbt shells by typing sbt chapter08b in different terminals, and launch four nodes as follows:

```
sbt:chapter08b>runMain example.countwords.App seed 2551
sbt:chapter08b>runMain example.countwords.App director 2555
sbt:chapter08b>runMain example.countwords.App aggregator 2556
```

## 8.5 Resilient job

This section shows the fault tolerance of the cluster when nodes fail. After you start the cluster and the master processes texts, you can kill all aggregator nodes and wait. Before you start a new aggregator, while no workers are available, the master collects every failure as lag, and nothing is lost. As soon as a new aggregator joins the cluster, the master can successfully delegate work again until the lag eventually reaches zero. Here is the output after shutting down all aggregator nodes:

```
[18:45:15,531] [DEBUG] - failed, adding text to lag 59    ❶
[18:45:15,540] [DEBUG] - tick, current lag 60             ❷
[18:45:15,540] [DEBUG] - failed, adding text to lag 56
[18:45:15,540] [DEBUG] - failed, adding text to lag 57
[18:45:15,540] [DEBUG] - failed, adding text to lag 58
```

❶ Result of a timeout after the master sends requests to workers

❷ Result when the master creates a new text

The lag is not ordered, which illustrates the essence of the indeterminacy of messaging.